

SREE SWIYA YARLAGADDA

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PERSONAL SUMMARY

Software engineer with experience building backend services, integrating relational databases, and developing well-documented RESTful APIs. Skilled in Python, FastAPI, SQL, and Git, with a strong foundation in object-oriented programming, data structures, and Agile practices. Eager to grow as a backend engineer and contribute to scalable, production-ready systems.

WORK EXPERIENCE

ML Research Assistant | Salvatori Computer Science Center, USC

Aug 2023 – May 2025

- Designed and maintained backend APIs using **Python, FastAPI, and Postgres** to handle real-time data retrieval for research workflows, improving query performance by **35%** and ensuring consistent uptime during high-demand periods.
- Built automated **data ingestion and transformation pipelines** with Python and SQL to streamline preparation of large datasets, cutting processing times by **50%** and enabling faster experimentation.
- Created internal documentation and API specs that allowed research and engineering teams to integrate services seamlessly, reducing onboarding time for new collaborators.
- Debugged and optimized backend services in production environments, reducing triage and recovery time by **40%** during live deployments.

Software Development Engineer | Shure, Hyderabad, India

Jan 2021 – Dec 2022

- Developed and deployed **RESTful APIs** using Python, Node.js, and Postgres to support configuration updates for over **30,000 connected devices**, reducing latency by **35%** and improving operational efficiency.
- Automated unit testing and implemented **CI/CD pipelines** with Git and Jenkins, improving release stability and reducing deployment errors by **25%**.
- Built **telemetry dashboards** and alerting workflows to monitor production systems, enabling faster detection of issues and minimizing downtime during global rollouts.
- Collaborated with cross-functional teams in **Agile sprints** to deliver stable, production-ready features on time and with full test coverage.

Engineering Intern | Tata Consultancy Services, Hyderabad, India

Jul 2020 - Dec 2020

- Created data pipelines in **Python, Postgres, and Redis** to support high-volume ad ranking systems, achieving top-K lookups in under **80ms** and improving accuracy for analytics teams.
 - Scaled batch and streaming ingestion workflows on **GCP**, improving throughput and maintaining SLA compliance during heavy traffic periods.
 - Assisted in backend API integration and performance tuning, supporting consistent system reliability during peak operations.
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EDUCATION

Master of Science, Computer Science (Scientists and Engineers)

January 2023 – May 2025

University of Southern California, Los Angeles, Relevant Coursework: Deep Learning, LLMs, Scalable Systems

PROJECTS

Stock Trading application: Built a full-stack trading platform (**Finnhub + Polygon.io APIs**) supporting 4,800+ concurrent users, with low-latency market data retrieval and secure trade execution. Implemented HTTPS and JWT authentication for compliance-grade security and seamless user experience. **Source:** [Github-Project](#)

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, Node.js, TypeScript

Databases and version control : PostgreSQL, MySQL, Redis, Git, GitHub, Postman

Core Development and Cloud: OOP, REST API design, unit testing, Agile collaboration, AWS (Lambda, EC2, S3), GCP